

A Cryptographic Infrastructure for Friend-to-Friend Choreographies

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A framework for

¹See [4, 5, 2, 1] for a formal model of local-first coordination.

A framework for

friend-to-friend [7, 8]

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coordination

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local-first [3, 6]¹

coordination

and its (forthcoming) implementation

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*To appear on
JLAMP 2026*

- ▶ Motivations for re-considering distributed computing
(and not being "obscured by clouds")

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- ▶ A new choreographic model
 - ▶ F2F-choreographies
 - ▶ Nearbybytes

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- ▶ Implementation aspects

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Blessing and curse of clouds

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- ▶ enabler of service-oriented computing/microservices (software-as-service)
- ▶ anytime-anywhere motto
- ▶ dynamic reconfiguration (services can be replaced if needed)
- ▶ reduced costs for maintenance, facilities, personnel, software, and hardware

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Where would you like to keep your data?
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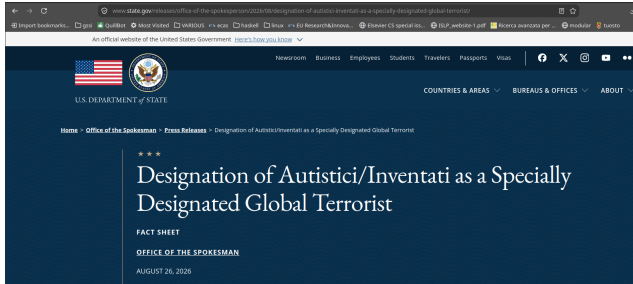
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Anytime, anywhere...really?

- ▶ *like during the AWS's outage on 25/11/2020*
- ▶ *or almost all Google services down on 14/12/2020*
- ▶ *DSL typical availability of 97%*
- ▶ *actually, some SLA have no lower bound [What Is the Digital Divide?]*

...and limited autonomy/sovereignty



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en.wikipedia.org/wiki/Autistici/Inventati

QuillBot Most Visited VARIOUS ecas haskell linux EU Research&Innova... Elsevier CS special iss...

as Monuments: Photograph a monument, help Wikipedia and win!

more

Autistici/Inventati

Article Talk

From Wikipedia, the free encyclopedia

"A/I" redirects here. For other uses, see [AI \(disambiguation\)](#).

This article **has an unclear citation** references used may be made clearer by using a different or consistent style of **citing** **footnoting**. (September 2026) ([Learn how to remove this message](#))

Autistici/Inventati (A/I) was an Italian **hacktivist** group that provided Internet communication services for groups opposed to **fascism**, **militarism**, **racism**, **sexism**, **homophobia** and **transphobia**.^[1]

Abbreviation	A/I
Established	2010s
Dissolved	6 September 2026
Type	Non-profit
Services	Encryption, web hosting, and mapping of

...and limited autonomy/sovereignty



en.wikipedia.org/wiki/Autistici/Inventati

Monuments: Photograph a monument, help Wikipedia and win!

Autistici/Inventati

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Perhaps some re-thinking is needed?

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- ▶ retained control over
 - ▶ data (i.e., ensure **confidentiality**)
 - ▶ computational resources
- ▶ had more "autonomy"

What if we go friend-to-friend too?

In Popescu, Crispo & Tanenbaum words

[...]Friends Team-Up and Beat the System

Abstract [...] while existing peer-to-peer architectures with similar aims attempt to build trust relationships on top of the basic, trust-agnostic, peer-to-peer overlay, Turtle takes the opposite approach, and builds its overlay on top of pre-existent trust relationships among its users. [...]

In Li & Dabek's words

F2F: reliable storage in open networks

Abstract [...] We argue that a node should choose its neighbors (the nodes with which it shares resources) based on existing social relationships instead of randomly. This approach provides incentives for nodes to cooperate and results in a more stable system which, in turn, reduces the cost of maintaining data. [...]

Local-first coordination
via
Friend-to-Friend Choreography

Our choreographic model

Syntax

$$\begin{aligned} F &::= A \rightsquigarrow B : t \\ &| F + F' \\ &| F ; F' \\ &| F \mid F' \\ &| \dots \end{aligned}$$

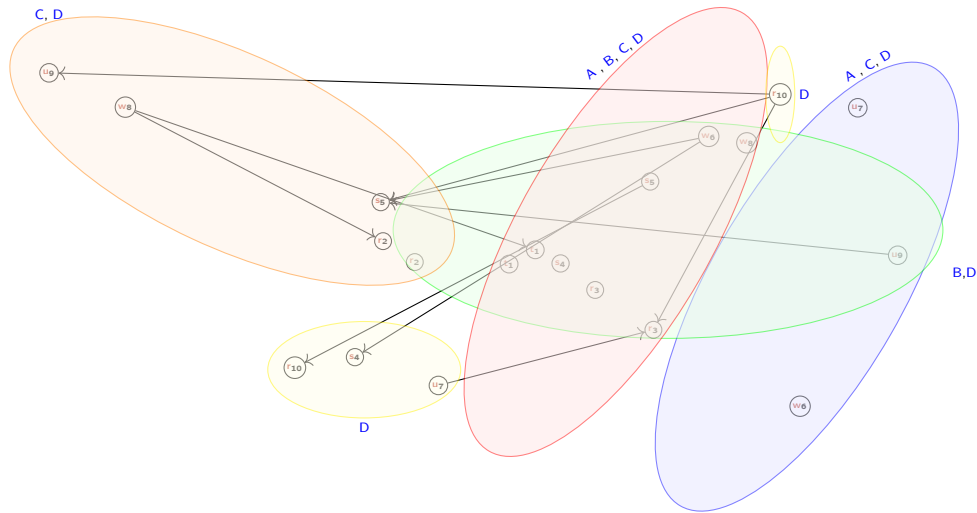
where

- ▶ B is a finite (possibly empty) set of A 's "buddies"
- ▶ t is a "type"

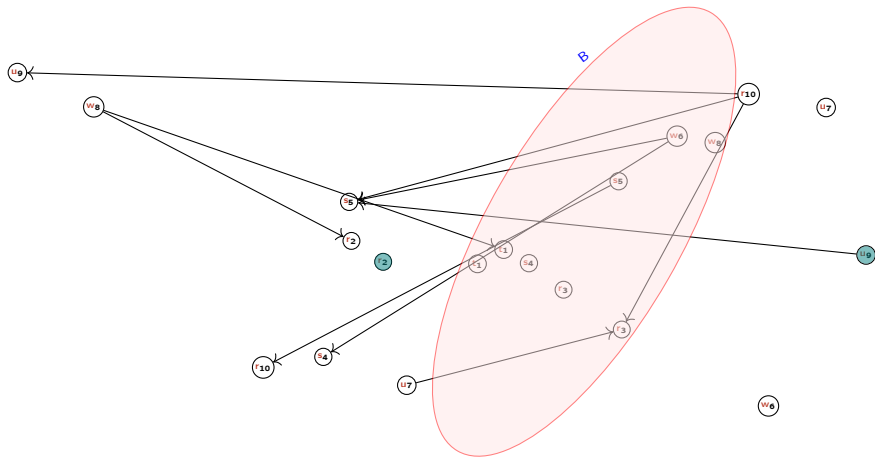
Semantics

$$\begin{aligned} \llbracket A \rightsquigarrow B : t \rrbracket &= \left\{ [A!B\langle t \rangle] \right\} \\ \llbracket F + F' \rrbracket &= \llbracket F \rrbracket \cup \llbracket F' \rrbracket \\ \llbracket F ; F' \rrbracket &= \{ r ;_{\varepsilon_r} r' \mid \dots \} \\ \llbracket F \mid F' \rrbracket &= \{ r ;_{\emptyset} r' \mid \dots \} \\ \llbracket \dots \rrbracket &= \dots \end{aligned}$$

The coordination model, informally



Replication



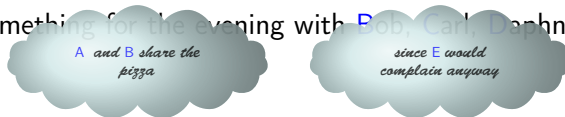
Having fun together

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Alice wants to arrange something for the evening with Bob, Carl, Daphne, and Evelin

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A's options are between staying home for a pizza or going out to a cinema

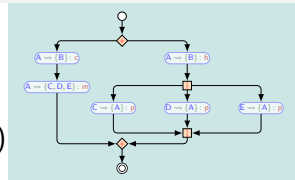
- ▶ if A opts for a cinema, B decides which movie; E should be oblivious of the title
- ▶ otherwise, deal is struck with whoever tells A which pizza they want

An F2F-choreography for fun

$$\begin{aligned} F = & \quad A \rightsquigarrow B : c; B \rightsquigarrow \{A, C, D\} : m \\ & + \quad A \rightsquigarrow B : h; (C \rightsquigarrow A : p \mid D \rightsquigarrow A : p \mid E \rightsquigarrow A : p) \end{aligned}$$

An F2F-choreography for fun

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Let's spell out some requirements:

- ▶ h and c may be public, but let's keep the rest confidential
- ▶ E should be able to know if A decided to go out or B choose the movie
- ▶ C , D , and E cannot choose a pizza until A makes her choice
- ▶ X 's choice of pizza is known only to X and A

Nearbytes and its cryptographic infrastructure

What is Nearbybytes?

What is Nearbybytes?

- ▶ set of protocols for confidential interactions combining
 - ▶ communication via "physical wiring"
 - ▶ with cryptography + "hubs" merging

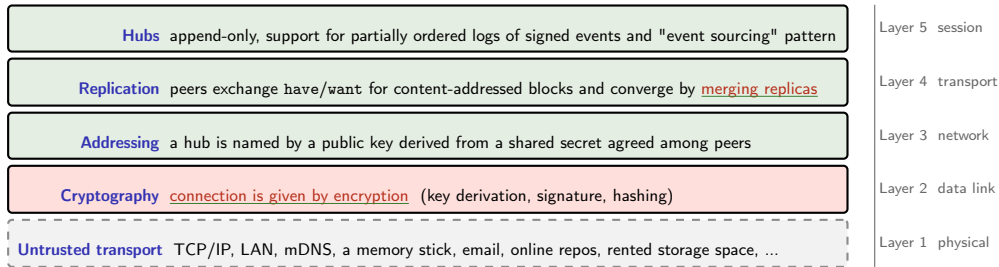
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- ▶ logical content-based addressing leveraging public-key cryptography
- ▶ data-related events form a partially ordered log

Nearbytes's stack

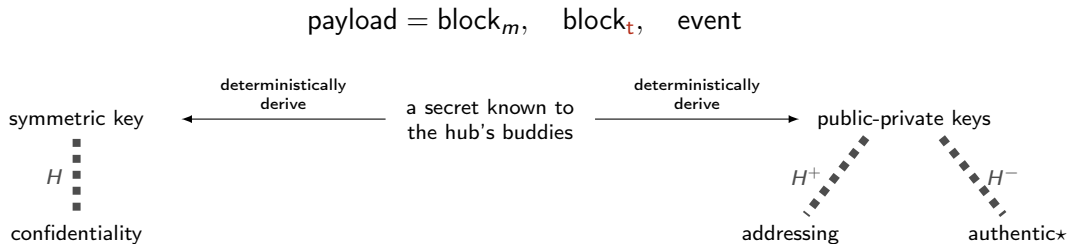


Nearbytes

Hush-hush (no key distribution)

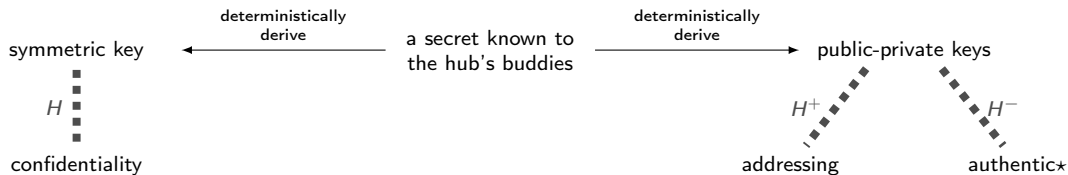
payload = block_m, block_t, event

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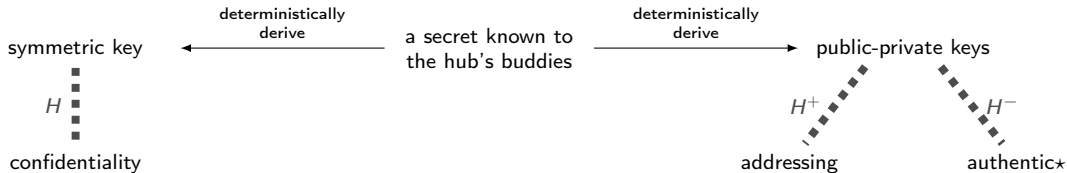
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$$\text{block}_m = \{ \text{msg} \}_H$$

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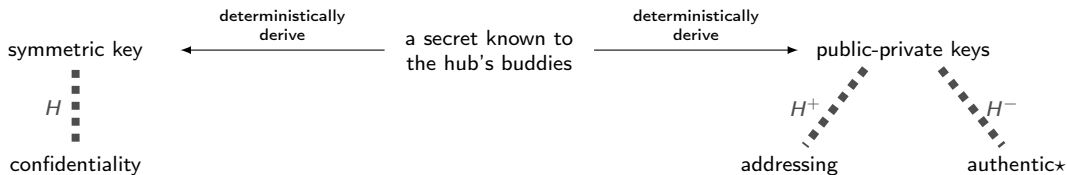
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block_m = { msg }_H
block_t = { type }_{H⁺}

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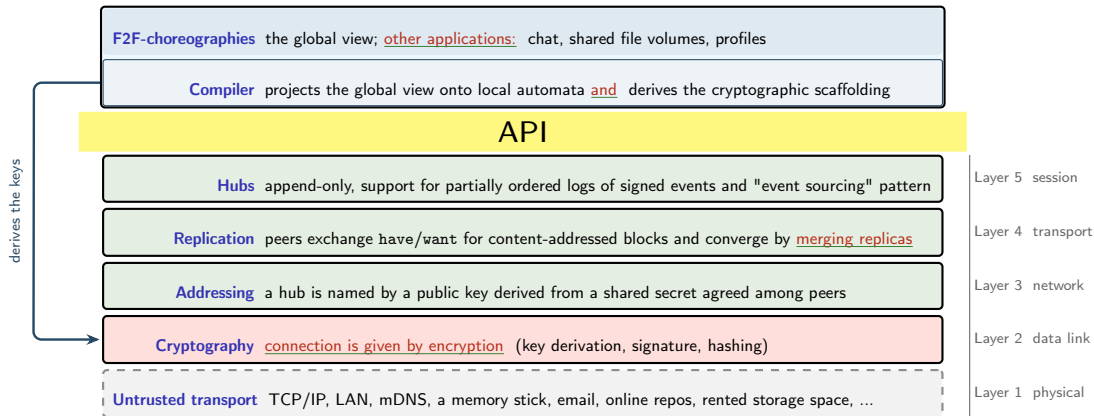
block_m = { msg }_H

block_t = { type }_{H⁺}

event = { hash(ev₁), ..., hash(ev_k), hash(block_m), hash(block_t) }_{H⁻}

F2F-choreographies & Nearbybytes

F2F-choreographies on top of Nearbybytes



Hubs' addressing

Buddies agree on a "master secret" M

Leverage buddies' public keys

$$\left\{ \begin{array}{lcl} H & = & M \cdot A^+ \cdot B^+ \cdot C^+ \cdot D^+ \cdot E^+ \\ H_1 & = & M \cdot A^+ \cdot B^+ \cdot C^+ \cdot D^+ \\ \dots & & \\ H_{14} & = & M \cdot A^+ \cdot B^+ \\ H_{15} & = & M \end{array} \right.$$

Projections

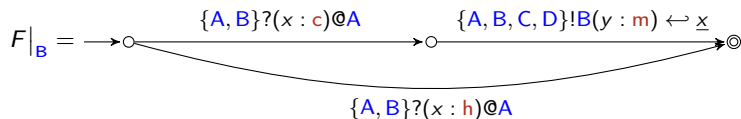
Deriving buddies' expected behaviour:

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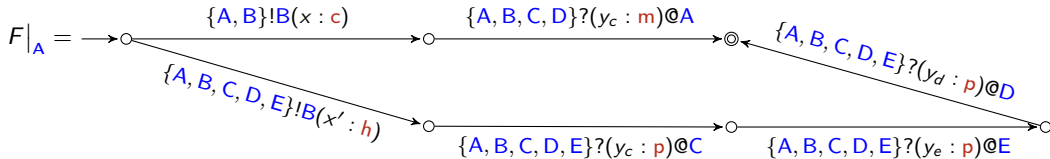
$$\begin{aligned} F = & \quad A \rightsquigarrow B : \textcolor{blue}{c}; B \rightsquigarrow \{A, \textcolor{blue}{C}, \textcolor{blue}{D}\} : \textcolor{blue}{m} \\ & + \quad A \rightsquigarrow B : \textcolor{blue}{h}; (C \rightsquigarrow A : \textcolor{blue}{p} \mid D \rightsquigarrow A : \textcolor{blue}{p} \mid E \rightsquigarrow A : \textcolor{blue}{p}) \end{aligned}$$



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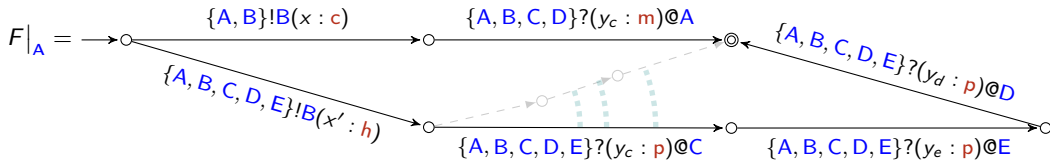
$$\left[A! \{A, B, C, D, E\} \langle h \rangle \right]$$

$$e = \{A \text{ 's address}\}_H, \{h\}_{H^+}, \{\text{hash}(\{A \text{ 's address}\}_H), \text{hash}(\{h\}_{H^+})\}_{H^-}$$

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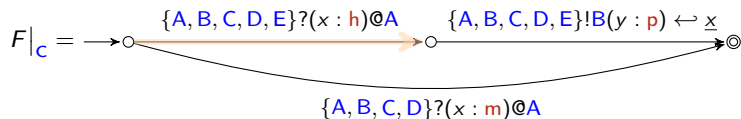
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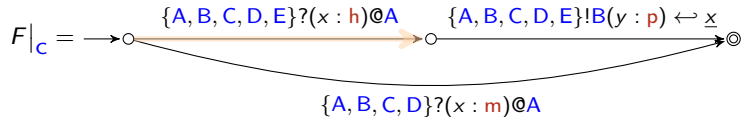
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$$\left[\begin{array}{c} A!\{A, B, C, D, E\}\langle \text{h} \rangle \\ \downarrow \\ C!\{A, B, C, D, E\}\langle \text{p} \rangle \end{array} \right]$$

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$$\{\text{margherita}\}_H, \{\text{p}\}_{H^+}, \{\text{hash}(e), \text{hash}(\{\text{margherita}\}_H), \text{hash}(\{\text{p}\}_{H^+})\}_{H^-}$$

Executing projections in Nearbybytes

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- ▶ Compile the automata to actual multithreaded code (e.g., Typescript or GoLang) and use `Nearbybytes` as a library

Wrapping-up

Future work

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- ▶ Consider specific notions of eventual consistency
- ▶ Extend *Nearbytes* with broadcast encryption schemes?

Thank you!

On "Obscured by clouds" I like

Bridges burning gladly
Merging with the shadows,
Flickering between the lines.
Stolen moments floating softly on the air,
Borne on wings of fire
and climbing higher.

Ancient bonds are breaking,
Moving on and changing sides.
Dreaming of a new day,
Cast aside the other way.

Magic visions stirring,
Kindled by and burning flames rise in her eyes.

The doorway stands ajar,
The walls that once were high.
Beyond the gilded cage,
Beyond the reach of ties.
The moment is at hand.

She breaks the golden band.

(Pink Floyd)

References I

- [1] Lucas Clorius, Florian Furbach, Roland Kuhn, Hernán Melgratti, Alceste Scalas, and Emilio Tuosto. Compositional Design, Implementation, and Verification of Swarms (Artifact). *Dagstuhl Artifacts Series*, 12(1):18:1–18:3, 2026.
- [2] Florian Furbach, Lucas Clorius, Roland Kuhn, Hernán C. Melgratti, Alceste Scalas, and Emilio Tuosto. Compositional Design, Implementation, and Verification of Swarms. In Robbert Krebbers and Alexandra Silva, editors, *40th European Conference on Object-Oriented Programming, ECOOP 2026, Brussels, Belgium, June 29 - July 3, 2026*, volume 372 of *LIPIcs*, pages 7:1–7:30. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2026.
- [3] Martin Kleppmann, Adam Wiggins, Peter van Hardenberg, and Mark McGranaghan. Local-first software: you own your data, in spite of the cloud. In *Proceedings of the 2019 ACM SIGPLAN International Symposium on New Ideas, New Paradigms, and Reflections on Programming and Software, Onward! 2019*, page 154–178, New York, NY, USA, 2019. Association for Computing Machinery.
- [4] Roland Kuhn, Hernán C. Melgratti, and Emilio Tuosto. Behavioural Types for Local-First Software. In Karim Ali and Guido Salvaneschi, editors, *37th European Conference on Object-Oriented Programming, ECOOP 2023, July 17-21, 2023, Seattle, Washington, United States*, volume 263 of *LIPIcs*, pages 15:1–15:28. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2023.
- [5] Roland Kuhn, Hernán C. Melgratti, and Emilio Tuosto. Behavioural Types for Local-First Software (Artifact). *Dagstuhl Artifacts Ser.*, 9(2):14:1–14:5, 2023.
- [6] Roland Kuhn and Wesley Reisz. Local-First Cooperation: Autonomy at the Edge, Secured by Crypto, 100% Available. <https://www.infoq.com/articles/local-first-cooperation>, May 2021. Last access on June 14, 2026.
- [7] Jinyang Li and Frank Dabek. F2f: reliable storage in open networks. <https://iptps06.cs.ucsb.edu/papers/Li-F2F06.pdf>.
- [8] Bogdan C. Popescu, Bruno Crispo, and Andrew S. Tanenbaum. Safe and private data sharing with turtle: Friends team-up and beat the system. In Bruce Christianson, Bruno Crispo, James A. Malcolm, and Michael Roe, editors, *Security Protocols*, pages 213–220, Berlin, Heidelberg, 2006. Springer Berlin Heidelberg.